



Image-Based Mushroom Edibility Classification with Deep Learning

DSC288R - Group 8

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Background

Problem

Mushroom identification is difficult and potentially dangerous. Many edible and poisonous mushrooms look visually similar, and misclassification can lead to severe poisoning or death.

- Manual identification requires expert knowledge.
- Existing tools rely on visual comparison and do not scale well.
- Real-world images vary in lighting, angle, background, and resolution.





Core Goal

Build an image-based deep learning model to classify mushrooms into:

Conditionally
Edible

Edible

Toxic
(combined Poisonous and Deadly)



Why This Problem Matters?

- Mushroom poisoning is a real public health risk.
- Non-experts cannot reliably distinguish species.
- Existing rule-based or manual tools:
 - Do not generalize well to diverse images
 - Cannot handle large-scale image variability
- Automation improves:
 - Accessibility
 - Speed
 - Scalability
 - Potential safety screening



Literature Survey

- **Past Approaches:**
 - Deep Learning CNN's: Resnet (He et al., 2016)
 - Vision Transformers (ViT): (Dosovitskiy et. al, 2021)
 - RandAugment (Cubuk et al., 2020)
- **Limitations**
 - Requires manual feature engineering
 - Hard to capture complex visual patterns
 - Poor generalization to uncontrolled environments
- **Gaps we address:**
 - 3 Class problem vs Binary
 - CNN vs. Transformers
 - Toxic Mushroom Edibility Metrics



Why ML/DL?

Strong Justification

Mushroom classification depends on:

- Subtle cap texture
- Gill structure
- Shape proportions

Transformers may capture:

- Global structure relationships
- Contextual spatial dependencies

Structured Comparison

These are hierarchical visual features
→ perfect for CNNs

Comparing between:

- Shallow CNN baseline (ResNet-18)
- Deeper CNN (ResNet-50)
- Transformer (Swin)

Dataset

Mushroom Classification Dataset (Kaggle)

<https://www.kaggle.com/datasets/zedsden/mushroom-classification-dataset>

102,711 images

4 classes: Conditionally Edible, Edible, Poisonous and Deadly



Feature Extraction & Data Pipeline

01

Data consolidation

02

Metadata enrichment

03

Exact duplicate removal (pHash)

04

Near duplicate removal

05

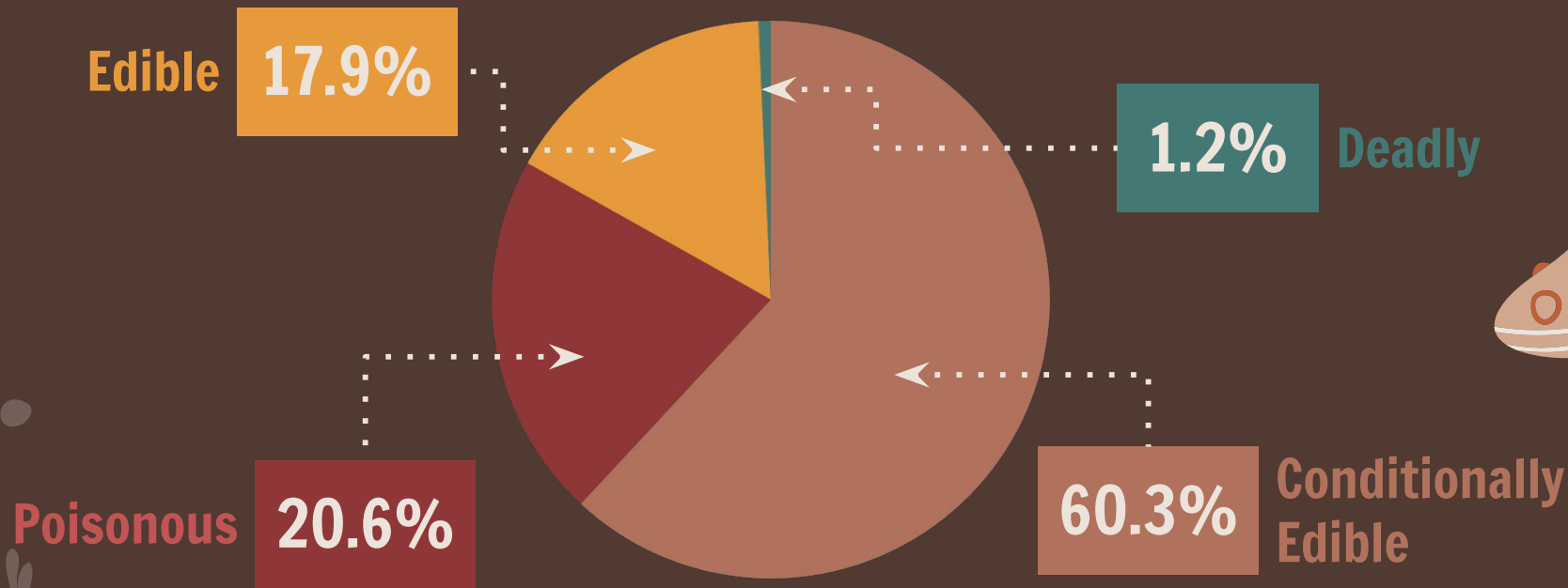
Standardization

06

Data augmentation

Original Classes

(94,631 images after removed duplicates)



EDA Insights

Key findings:

- Severe class imbalance
- Large resolution variability
- Extreme aspect ratios in minority images
- Duplicate clusters across species folders

Interpretation:

- Combine **poisonous** and **deadly** classes to one single **toxic** class to improve minority class performance.
- Resizing and augmentation were necessary to reduce bias.

Updated Classes

(after merging poisonous class and deadly class)

Edible

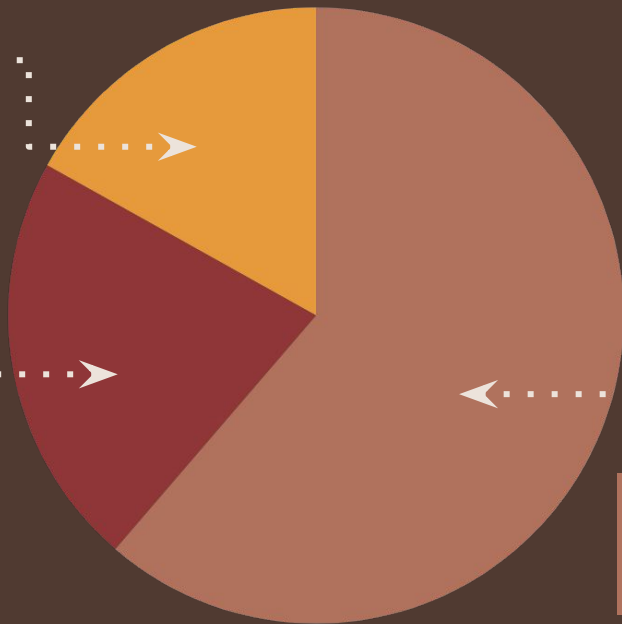
16.9%

Toxic

21.8%

60.3%

Conditionally
Edible



Augmentation Strategy

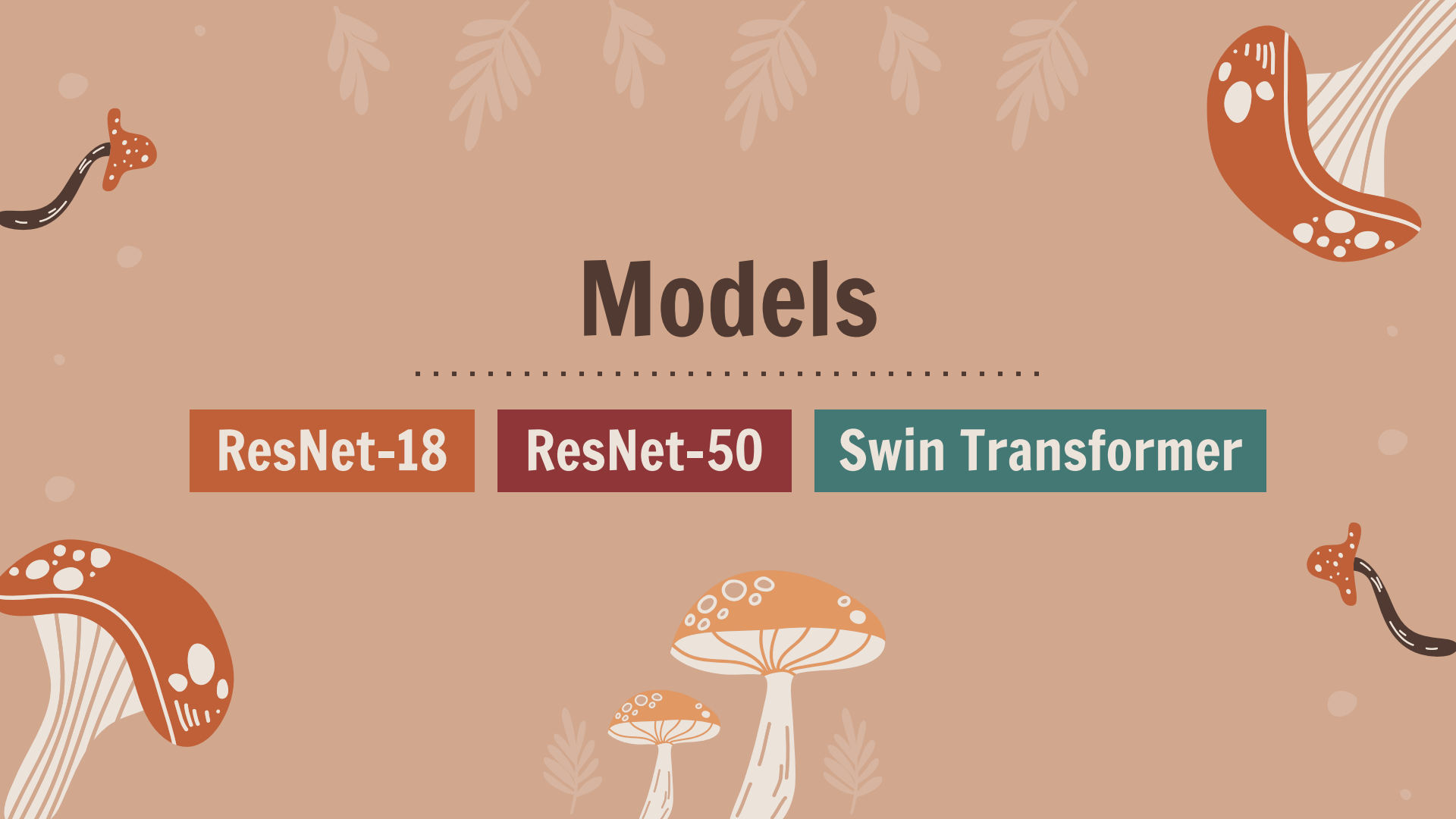
Applied only to training set

- Random horizontal flip
- Rotation
- Color jitter

Purpose

- Simulate real-world lighting variation
- Improve invariance
- Prevent overfitting to background



The background is a solid light brown color. It is decorated with several stylized illustrations of mushrooms and leaves. In the top left, there is a small red mushroom with white spots and a black stem. In the top right, there is a large red mushroom with white spots and a white stem. In the bottom left, there is a large red mushroom with white spots and a white stem. In the bottom center, there are two smaller red mushrooms with white spots and white stems. In the bottom right, there is a small red mushroom with white spots and a black stem. There are also several light brown leaves scattered across the background.

Models

ResNet-18

ResNet-50

Swin Transformer

ResNet-18

Convolutional Neural Network (CNN)

Model Used: ResNet-18 (Pretrained on ImageNet)

- Pretrained weights: ImageNet (transfer learning)
- Input size: 224×224 RGB
- Final layer modified to 3 classes
- Optimizer: AdamW ($\text{lr} = 3\text{e-}4$)
- Early stopping on validation loss

Why ResNet-18?

- Residual (skip) connections mitigate vanishing gradients
- ~11 million parameters (computationally efficient)
- Strong and stable baseline for image classification
- Suitable for medium-sized datasets



Feature Hierarchy

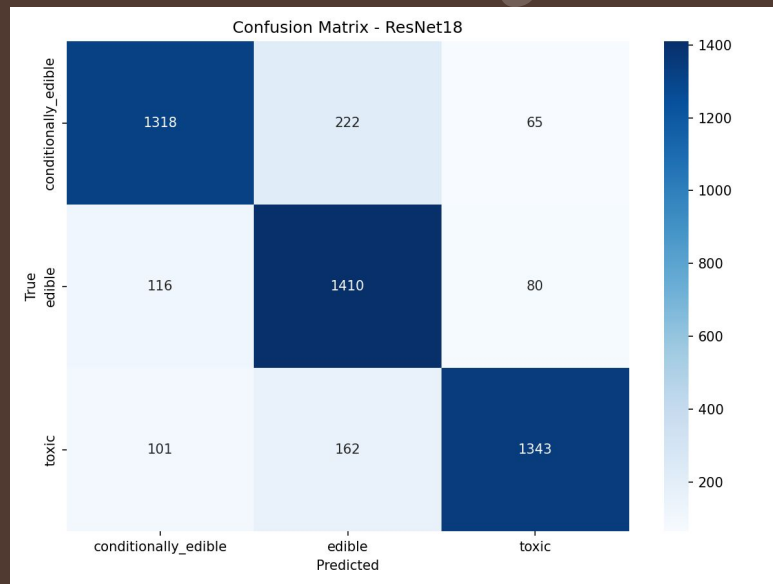
Stage	Output Size	Channels
Input	224×224	3
Conv1 (stem layer)	112×112	64
Layer1	56×56	64
Layer2	28×28	128
Layer3	14×14	256
Layer4	7×7	512
AvgPool + FC	1×1	3

Observations & Results

Overall Accuracy: 84.51%

Classification report:

	precision	recall	f1-score	support
conditionally_edible	0.8586	0.8212	0.8395	1605
edible	0.7860	0.8780	0.8294	1606
toxic	0.9026	0.8362	0.8681	1606
accuracy			0.8451	4817
macro avg	0.8490	0.8451	0.8457	4817
weighted avg	0.8490	0.8451	0.8457	4817



ResNet-50

Deep Convolutional Neural Network (CNN)

Model Used

- Pretrained weights, Input size and Final layer same as ResNet-18
- Optimizer: AdamW ($\text{lr} = 1\text{e-}3$)
- Early stopping on validation loss

Why ResNet-50?

- Deeper architecture and better hierarchical feature extraction (50 layers)
- Bottleneck residual blocks improve feature learning
- Stronger representation capacity than ResNet-18
- ~25M parameters (more expressive model)
- Captures more complex texture & structural patterns
- Suitable for fine-grained visual differences (mushroom classes)



Observations & Results

Overall Accuracy: 88.42%

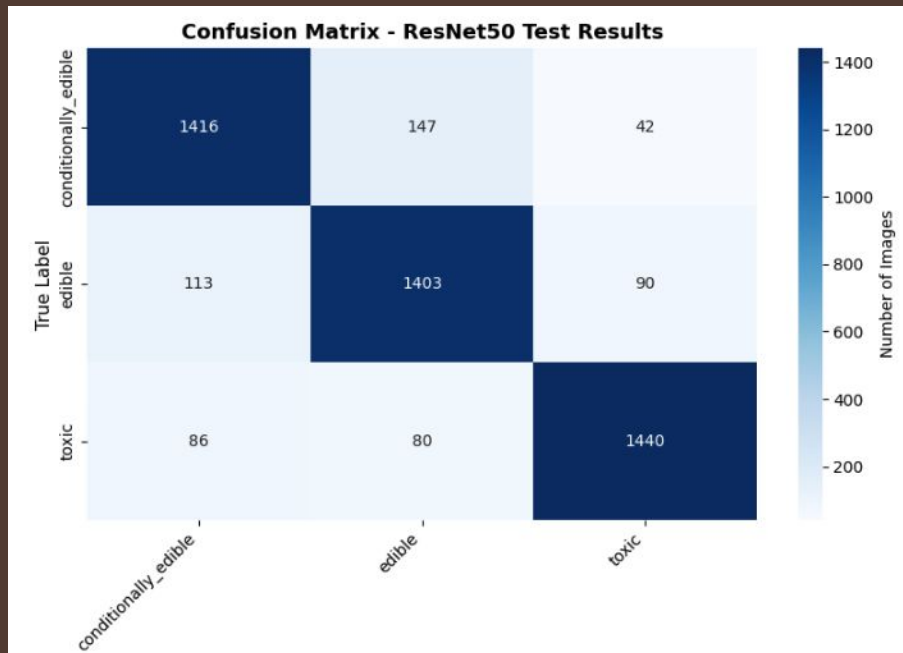
ResNet50 Test Results:

Total test images: 4817
Correct predictions: 4259
Test Accuracy: 88.42%

Per-Class Accuracy:

conditionally_edible: 1416/1605 = 88.22%
edible : 1403/1606 = 87.36%
toxic : 1440/1606 = 89.66%

AVERAGE : = 88.42%



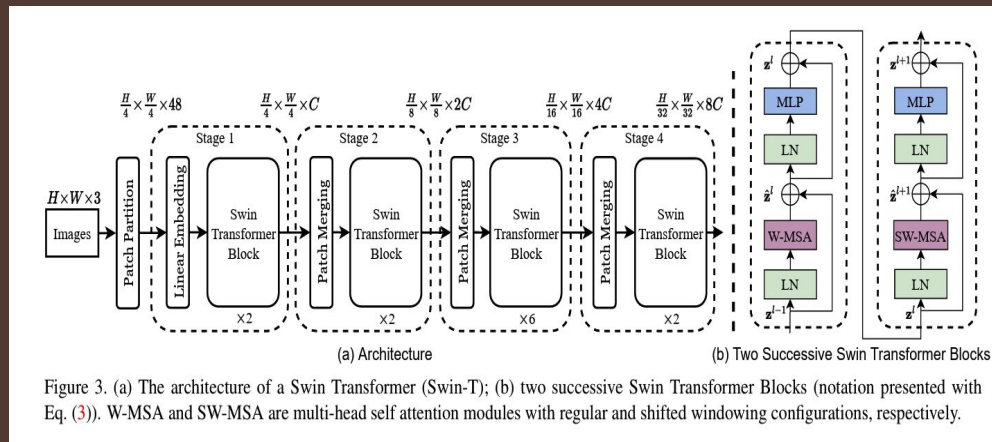
SWIN Transformer Tiny

What is SWIN Transformer?

- Vision Transformer Introduced by Liu et al. (2021)
- Designed to bring transformer efficiency to computer vision
- "SWIN" = Shifted Windows
- SWIN processes images hierarchically (like CNNs do) making it faster

Why Did We Choose It?

- Mushroom class needs fine details and overall image structure information
- ~28M parameters which is a good Balance of power and efficiency



Observations & Results

Overall Accuracy: 90.47%

Per-Class Performance:

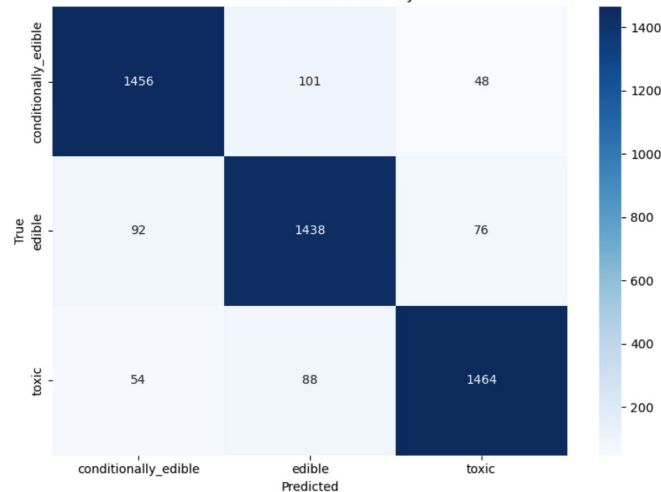
- **Toxic Class Performance (Highest Priority)**
 - 91.16% Recall, 92.19% Precision
 - 88 out of 1606 Toxic → Edible Misclassifications
 - This error occurs 5.5% of the time
- **Conditionally Edible Performance**
 - 90.89% Precision, 90.72% Recall
- **Edible Performance**
 - 88.38% Precision, 89.54% Recall

Overall Accuracy: 0.9047

Balanced Accuracy: 0.9047

	precision	recall	f1-score	support
conditionally_edible	0.9089	0.9072	0.9080	1605
edible	0.8838	0.8954	0.8896	1606
toxic	0.9219	0.9116	0.9167	1606
accuracy			0.9047	4817
macro avg	0.9049	0.9047	0.9048	4817
weighted avg	0.9049	0.9047	0.9048	4817

Confusion Matrix - Swin Tiny



Next Steps before Final Report

Implement Grad-CAM (as suggested by TA) to verify model focuses on features of mushroom and not background of image

Improve fine tuning and data augmentation techniques to improve performance of Toxic class and per-class recall

Implementing ensemble methods or integrating confidence thresholds and reject option for low-confidence predictions

Improve fine-tuning strategies (layer unfreezing schedule, learning rate tuning)



References

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<https://www.kaggle.com/datasets/zedsden/mushroom-classification-dataset>





Thank You!

